# Gabriel MARIADASS | Paris, France

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## SUMMARY

M.Sc. MVA at ENS Paris-Saclay and M.Eng. at École Nationale des Ponts et Chaussées, combining applied ML engineering at Dassault Systèmes and Banque de France with research in uncertainty-aware ML at NTU Singapore. Interested in a CIFRE PhD or an AI Engineering role in industry, startup or scale-up. Available from October 2026.

## EDUCATION

**ENS Paris-Saclay, France** *2025 - 2026*
*M.Sc. Mathematics, Vision, and Machine Learning (MVA)*

- **Academic average:** 17.19/20 (with Highest Honors)
- **Key courses**: Deep Learning, Advanced Learning for Text and Graph Data (ALTEGRAD), Graphs in Machine Learning, Bayesian Machine Learning, Convex Optimization for Machine Learning, Computational Statistics, Machine Learning for Time Series

**École Nationale des Ponts et Chaussées, France** *2022 - 2026*
*M.Eng. - Mathematical and Computer Engineering (IMI)*
Final year completed at ENS Paris-Saclay (MVA)

**Lycée Henri-IV, France** *2020 - 2022*
*CPGE - PCSI/PC**

## RESEARCH EXPERIENCE

**LPSM, France & NTU, Singapore** *04/2026 - 08/2026*
*Research Intern - Possibilistic Variational Inference*
**Supervisors**: Badr-Eddine Chérief-Abdellatif (CNRS, LPSM) and Jérémie Houssineau (NTU)

- Reproduced published image-classification results (LeNet-5, Fashion-MNIST) to build a validated baseline for new methods
- Identified why the model's uncertainty estimates were unreliable on four test cases and traced it to the choice of model family
- Designed learned uncertainty structures across four model families, improving accuracy, NLL, and calibration
- Built on an existing SAM connection to derive an adaptive, automatically learned perturbation radius

## PROFESSIONAL EXPERIENCE

**Dassault Systèmes, France** *01/2025 - 07/2025*
*ML Engineer - Deep Learning & GNN (Molecular Retrosynthesis)*

- Migrated GNN retrosynthesis pipeline from DGL to PyTorch Geometric, cutting CPU inference time by 38%
- Improved model accuracy by up to 2 percentage points via Bayesian hyperparameter optimization
- Explored quantization (dynamic/static) for inference speedup and refactored the preprocessing pipeline for maintainability

**Banque de France, France** *09/2024 - 12/2024*
*Data Scientist - On-site Credit Institution Control Delegation*

- Built neural network credit-scoring models with AUC 0.83 on 700K loan contracts, raising Gini +0.14 vs. internal IRB models
- Led PD/LGD/EAD risk-parameter analysis and time-series studies for a live ECB-mandated audit of a major French bank

## TECHNICAL SKILLS

- **DL & ML**: PyTorch, PyTorch Geometric, DGL, scikit-learn, XGBoost/LightGBM, NumPy, Pandas, GNNs, Deep Learning, Variational Inference, Bayesian Optimization, Retrieval, Reranking, Text Embeddings, LLM Fine-Tuning (LoRA)
- **MLOps & Deployment (see** price-api in portfolio**):** Docker, FastAPI, MLflow, CI/CD (GitHub Actions), Cloud (GCP), pytest, Git
- **Programming**: Python, C++, SQL
- **Mathematics**: Convex Optimization, Bayesian Learning, Time Series, Information Theory, Possibility Theory, MCMC, PAC-Bayes, Extreme Value Theory

## LANGUAGES & INTERESTS

- **Languages**: French (native), English (C1), Spanish (B1)
- **Interests**: Muay Thai / Kickboxing, Running, Weightlifting, Nutrition, AI Tech Watch, Tech Entrepreneurship, Reading, Travel